

****Practical: 01**

Title: Installation and setup of java development kit (JDK), setup android SDK, setup eclipse IDE, setup android development tools (ADT) plugins, create android virtual device.

Step 1: Installation of Java Development Kit (JDK)

1. Open a web browser and download the latest JDK from the official Oracle website.

Step 2: Select the Appropriate Version

As of 2025, the latest stable versions are JDK 23 (SE) and JDK 21 (LTS). Select the compatible version as per your operating system (Windows, Mac or Linux)

Step 3: License and Agreement

Go through all the License and Agreement before downloading (from Oracle website), it will not ask if you'll download it from OpenJDK website.

Install JDK on Windows

Follow the below steps to install JDK on Windows environment i.e. Windows 7, Windows 8, Windows 8.1, Windows 10 and Windows 11.

Step 1: Run the Java Development Kit (JDK) Installer

Locate the downloaded .exe file (e.g. jdk-23-windows-x64_bin.exe) and make the double click to begin the Installation process. Follow the installation wizard prompts to complete the installation process.

Step 2: Setup the Environment Variables

Step 2.1: To set the Environment Variables, you need to search Environment Variables in the Task Bar and click on "Edit the system environment variables".

Step 2.2: Under the Advanced section, Click on "Environment Variables".

Step 2.3: Under System variables, select the "Path" variable and click on "Edit". Click on "New" then paste the Path Address i.e. C:\Program Files\Java\jdk-

{YOUR_JDK_VERSION}\bin. Click on "OK".

Step 3: Check the Java Version

Open Command Prompt and enter the following commands:

```
java -version
```

```
javac -version
```

2. Steps to Install Android Studio on Windows

Step 1: Head over to this link to get the Android Studio executable or zip file.

Step 2: Click on the Download Android Studio Button.

Click on the "I have read and agree with the above terms and conditions" checkbox followed

step 3: After the downloading has finished, open the file from downloads and run it. It will prompt the following dialog box.

Click on next.

Step 5: Once " **Finish** " is clicked, it will ask whether the previous settings need to be imported [if the android studio had been installed earlier], or not. It is better to choose the 'Don't import Settings option'.

Step 7: After it has found the SDK components, it will redirect to the Welcome dialog box.

Step 8: Now it is time to download the SDK components.

The Android Studio has been successfully configured. Now it's time to launch and build apps. Click on the Finish button to launch it.

3. Steps to Setup Eclipse IDE and setup android development tools (ADT) plugins for Android App Development

Step 1: Open Eclipse IDE.

Step 2: Select Help, Click on "Install New Software"

Step 3: Type "<https://dl-ssl.google.com/android/eclipse/>" in the "Work With" section and click on Add. Further, a new Dialog box will appear, type Name - ADT Repository and Click on Add.

Step 4: You can see Developer Tools under Name - tick the box, then click Next. A dialog box will appear, click on Next and then click on Finish. After that, Installation will begin.

Step 5: After the installation is completed, Eclipse will be restarted. After the restart, a dialog box will appear for setting up the Preferences. Click on Open Preferences then Click on Proceed. If the dialog box does not appear then go to Eclipse -> Window -> Preferences.

Step 8: After installation is completed, Go to Eclipse then Select Window then Click on Android Virtual Device Manager. A dialog box will appear, Select existing AVD and Click on Edit.

3. Create Android Virtual Device(AVD)

4. **Step 1:** Go to **Tools > AVD Manager**.

Step 2: **Now click on** Create Virtual Device.

Step 3: A pop-up window will be there and here we select the category Phone because we are creating android app for mobile and select the model of mobile phone we want to install.

Step 5: **Click the finish button to complete the installation.**

Step 5: **Click the finish button to complete the installation.** **Step 6:** Now we can select the virtual device we want to run as emulator can click on the **run** icon.

Step 7: Finally our virtual device is ready to run our android

****Practical: 02 Title: Create “Hello World” application. That will display “Hello World” in the middle of the screen using TextView Widget in the red color.**

Step 1: Open Android Studio

1. Click on the Start Menu (Windows).

2. Search for Android Studio.

3. Click on Android Studio to open it.

4. Wait until the Welcome to Android Studio screen appears.

Step 2: Create a New Project

1. On the Welcome screen, click on New Project.

2. A list of project templates will be displayed.

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Step 3: Select Project Template

1. Select Empty Activity.

2. Click on the Next button.

Step 4: Configure Your Project

1. Name: Enter the project name Example: HelloWorldApp

2. Package Name: Auto-generated (leave as it is).

3. Save Location: Default location (leave as it is).

4. Language: Select Java.

5. Minimum SDK: Select API 21: Android 5.0 (Lollipop) or as required.

6. Click on the Finish button.

Step 5: Project Creation

1. Android Studio will start building the project.

2. Wait until Gradle Build Finished message appears at the bottom.

3. Your project is now successfully created.

Step 6: Open Layout File

1. Go to app → res → layout.

2. Open activity_main.xml.

3. Add or modify the TextView code as required.

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Android Layout File (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?> <LinearLayout  
xmlns:android="http://schemas.android.com/apk/res/android"  
android:layout_width="match_parent" android:layout_height="match_parent"  
android:gravity="center" android:orientation="vertical">
```

```
<TextView android:layout_width="wrap_content" android:layout_height="wrap_content"  
android:text="Hello World" android:textColor="#00FF00" android:background="#BFBF0E"  
android:textSize="80sp" /> </LinearLayout>
```

****Practical: 03 Title:Create Registration page to demonstration of Basic widgets available in android.**

Step-by-Step Instructions:

Step 1: Open Android Studio

- o Click on the Start Menu (Windows).
- o Search for Android Studio.
- o Click on Android Studio to open it.
- o Wait until the Welcome to Android Studio screen appears.

Step 2: Create a New Project

- o On the Welcome screen, click on **New Project**.
- o A list of project templates will be displayed.

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Step 3: Select Project Template

- o Select Empty Activity.
- o Click on the Next button.

Step 4: Configure Your Project

- o Name: Enter the project name Example: HelloWorldApp
- o Package Name: Auto-generated (leave as it is).
- o Save Location: Default location (leave as it is).
- o Language: Select Java.
- o Minimum SDK: Select API 21: Android 5.0 (Lollipop) or as required.
- o Click on the Finish button.

Step 5: Project Creation

- o Android Studio will start building the project.
- o Wait until **Gradle Build Finished** message appears at the bottom.
- o Your project is now successfully created.

Step 6: Open Layout File

- o Go to **app → res → layout**.
- o Open **activity_main.xml**.
- o Add or modify the **TextView code** as required.

Android Layout File (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?> <RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent" android:layout_height="match_parent"
android:padding="16dp"> <!-- Title --> <TextView android:id="@+id/tvTitle"
android:layout_width="wrap_content" android:layout_height="wrap_content"
android:text="Registration Form" android:textSize="24sp" android:textStyle="bold"
android:layout_centerHorizontal="true"/> Page No -

<!-- Name Label --> <TextView android:id="@+id/tvName" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Name" android:textSize="20sp"
android:textStyle="bold" android:layout_below="@id/tvTitle" android:layout_marginTop="15dp"/>
<!-- Name Input --> <EditText android:id="@+id/etName" android:layout_width="match_parent"
android:layout_height="wrap_content" android:layout_below="@id/tvName"/> <!-- Address Label --
> <TextView android:id="@+id/tvAddress" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Address" android:textSize="19sp"
android:textStyle="bold" android:layout_below="@id/etName" android:layout_marginTop="10dp"/>
<!-- Address Input --> <EditText android:id="@+id/etAddress" android:layout_width="match_parent"
android:layout_height="wrap_content" android:layout_below="@id/tvAddress" /> <!-- City Label -->
<TextView android:id="@+id/tvCity" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="City" android:textSize="18sp" Page No -

android:textStyle="bold" android:layout_below="@id/etAddress"
android:layout_marginTop="10dp"/> <!-- City Input --> <EditText android:id="@+id/etCity"
android:layout_width="match_parent" android:layout_height="wrap_content"
android:layout_below="@id/tvCity"/> <!-- Submit Button --> <Button android:id="@+id/btnSubmit"
android:layout_width="match_parent" android:layout_height="wrap_content"
android:text="Submit" android:textSize="19sp" android:textStyle="bold"
android:layout_below="@id/etCity" android:layout_marginTop="20dp"/> </RelativeLayout>
```

****Practical: 04 Title: Create sample application with login module.(Check username and password)
On successful login, Change TextView “Login Successful”. And on failing login, alert user using Toast
“Login fail”.**

Android Layout File (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?> <LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent" android:layout_height="match_parent"
android:orientation="vertical" android:padding="20dp"> <!-- Title --> <TextView
android:id="@+id/tvStatus" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Login Page" android:textSize="24sp"
android:textStyle="bold" android:layout_gravity="center" android:paddingBottom="20dp"/> <!--
Username --> <EditText android:id="@+id/etUsername" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Enter Username"/> <!-- Password --> <EditText
android:id="@+id/etPassword" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Enter Password"
android:inputType="textPassword" android:layout_marginTop="10dp"/> Page No -
```

```
<!-- Login Button --> <Button android:id="@+id/btnLogin" android:layout_width="match_parent"
android:layout_height="wrap_content" android:text="Login" android:layout_marginTop="20dp"/>
</LinearLayout>
```

****Practical: 05 Title: Create an application for demonstration of Scroll view in android.**

Android Layout File (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?> <ScrollView
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent" android:layout_height="match_parent"> <LinearLayout
android:layout_width="match_parent" android:layout_height="wrap_content"
android:orientation="vertical" android:padding="16dp"> Page No -

<!-- Multiple TextViews to demonstrate scrolling --> <TextView
android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Item
1" android:textSize="24sp" android:padding="10dp"/> <TextView
android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Item
2" android:textSize="24sp" android:padding="10dp"/> <TextView
android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Item
3" android:textSize="24sp" android:padding="10dp"/> <!-- Add more TextViews to exceed screen
height --> <TextView android:layout_width="match_parent" android:layout_height="wrap_content"
android:text="Item 4" android:textSize="24sp" android:padding="10dp"/> <TextView
android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Item
5" android:textSize="24sp" android:padding="10dp"/> <TextView
android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Item
6" android:textSize="24sp" android:padding="10dp"/> Page No -

<TextView android:layout_width="match_parent" android:layout_height="wrap_content"
android:text="Item 7" android:textSize="24sp" android:padding="10dp"/> <TextView
android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Item
8" android:textSize="24sp" android:padding="10dp"/> <TextView
android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Item
9" android:textSize="24sp" android:padding="10dp"/> <TextView
android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Item
10" android:textSize="24sp" android:padding="10dp"/> <TextView
android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Item
11" android:textSize="24sp" android:padding="10dp"/> <TextView
android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Item
12" android:textSize="24sp" android:padding="10dp"/> </LinearLayout> </ScrollView>
```

****Practical: 06Title: Create login application where you will have to validate username and passwords till the username and password is not validated, login button should remain disabled.**

Android Layout File (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
```

```
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="24dp">
<EditText
    android:id="@+id/etUsername"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Username"
    android:inputType="text" />
<EditText
    android:id="@+id/etPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Password"
    android:inputType="textPassword"
    android:layout_marginTop="16dp" />
<Button
    android:id="@+id/btnLogin"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Login"
    android:enabled="false"
    android:layout_marginTop="24dp"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.loginapp;

import android.os.Bundle;

import android.text.Editable;

import android.text.TextWatcher;
```

```
import android.widget.Button;

import android.widget.EditText; Page No -

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etUsername, etPassword;

    Button btnLogin;

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        etUsername = findViewById(R.id.etUsername);

        etPassword = findViewById(R.id.etPassword);

        btnLogin = findViewById(R.id.btnLogin);

        TextWatcher textWatcher = new TextWatcher() {

            @Override

            public void beforeTextChanged(CharSequence s, int start, int count, int after) { }

            @Override

            public void onTextChanged(CharSequence s, int start, int before, int count) {

                validateInputs();

            }

            @Override

            public void afterTextChanged(Editable s) { }

        };

        etUsername.addTextChangedListener(textWatcher);

        etPassword.addTextChangedListener(textWatcher);

    }

    private void validateInputs() {

        String username = etUsername.getText().toString().trim();

        String password = etPassword.getText().toString().trim();

        boolean isValid = username.length() >= 4 && password.length() >= 6;

        btnLogin.setEnabled(isValid);

    }

}
```


} Page No -

}

****Practical: 07 Title: Create an application for calculator**

Activity_main.xml (res/layout/activity_main.xml)

<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

android:layout_width="match_parent" Page No -

android:layout_height="match_parent"

android:orientation="vertical"

android:background="#F5F5F5"

android:padding="12dp">

<!-- Display -->

<TextView

android:id="@+id/tvDisplay"

android:layout_width="match_parent"

android:layout_height="80dp"

android:background="FFFFFF"

android:textColor="000000"

android:textSize="28sp"

android:gravity="center_vertical|end"

android:padding="16dp"

android:text="125+35 = 160" />

<GridLayout

android:layout_width="match_parent"

android:layout_height="wrap_content"

android:columnCount="4"

android:layout_marginTop="12dp"> Page No -

<!-- Row 1 -->

<Button android:text="C" style="@style/BtnClearLight" android:onClick="clear"/>

```

<Button android:text="/" style="@style/BtnOpLight" android:onClick="input"/>
<Button android:text="x" style="@style/BtnOpLight" android:onClick="input"/>
<Button android:text="-" style="@style/BtnOpLight" android:onClick="input"/>
<!-- Row 2 -->
<Button android:text="7" style="@style/BtnNumLight" android:onClick="input"/>
<Button android:text="8" style="@style/BtnNumLight" android:onClick="input"/>
<Button android:text="9" style="@style/BtnNumLight" android:onClick="input"/>
<Button android:text="+" style="@style/BtnOpLight" android:onClick="input"/>
<!-- Row 3 -->
<Button android:text="4" style="@style/BtnNumLight" android:onClick="input"/>
<Button android:text="5" style="@style/BtnNumLight" android:onClick="input"/>
<Button android:text="6" style="@style/BtnNumLight" android:onClick="input"/>
<!-- Row 4 -->
<Button android:text="1" style="@style/BtnNumLight" android:onClick="input"/>
<Button android:text="2" style="@style/BtnNumLight" android:onClick="input"/>
<Button android:text="3" style="@style/BtnNumLight" android:onClick="input"/>
<!-- Row 5 -->
<Button android:text="0" style="@style/BtnNumLight" android:onClick="input"/> Page No -
<Button android:text="." style="@style/BtnNumLight" android:onClick="input"/>
<Button
android:text="="
android:layout_columnSpan="2"
style="@style/BtnEqualLight"
android:onClick="calculate"/>
</GridLayout>
</LinearLayout>

```

styles.xml (res/values/styles.xml)

```

<resources>
<style name="BtnNumLight">
<item name="android:layout_width">0dp</item>
<item name="android:layout_height">70dp</item>

```

<item name="android:layout_columnWeight">1</item>

<item name="android:textSize">20sp</item>

<item name="android:background">#FFFFFF</item>

<item name="android:textColor">#000000</item>

<item name="android:layout_margin">4dp</item>

<item name="android:elevation">4dp</item>

</style> Page No -

<style name="BtnOpLight" parent="BtnNumLight">

<item name="android:textColor">#1976D2</item>

</style>

<style name="BtnClearLight" parent="BtnNumLight">

<item name="android:background">#E53935</item>

<item name="android:textColor">#FFFFFF</item>

</style>

<style name="BtnEqualLight" parent="BtnNumLight">

<item name="android:background">#43A047</item>

<item name="android:textColor">#FFFFFF</item>

</style>

</resources>

MainActivity.java(java/com/example/calculator/MainActivity.java)

package com.example.calculator;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

import android.view.View;

import android.widget.Button;

import android.widget.TextView;

public class MainActivity extends AppCompatActivity { Page No -

TextView tvDisplay;

String expression = "";

@Override

```

protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    tvDisplay = findViewById(R.id.tvDisplay); }

    public void input(View view) {
        Button b = (Button) view;
        expression += b.getText().toString();
        tvDisplay.setText(expression); }

    public void clear(View view) {
        expression = "";
        tvDisplay.setText("");
    }

    public void calculate(View view) {
        try {
            expression = expression.replace("x", "*");
            double result = eval(expression);
            tvDisplay.setText(expression + " = " + result); Page No -

```

```

expression = "";
        } catch (Exception e) {
            tvDisplay.setText("Error");
            expression = "";
        }
    }
}

```

// Simple evaluator

```

private double eval(String exp) {
    return new Object() {
        int pos = -1, ch;
        void nextChar() {
            ch = (++pos < exp.length()) ? exp.charAt(pos) : -1;
        }
        boolean eat(int charToEat) {

```

```

while (ch == ' ') nextChar();
if (ch == charToEat) {
    nextChar();
    return true; }
return false;
}

double parse() { nextChar(); Page No -

```

```

double x = parseExpression();
return x; }

double parseExpression() {
    double x = parseTerm();
    for (;;) {
        if (eat('+')) x += parseTerm();
        else if (eat('-')) x -= parseTerm();
        else return x;
    }
}

double parseTerm() {
    double x = parseFactor();
    for (;;) {
        if (eat('*')) x *= parseFactor();
        else if (eat('/')) x /= parseFactor();
        else return x;
    }
}

double parseFactor() {
    if (eat('+')) return parseFactor();
    if (eat('-')) return -parseFactor();
    double x; int startPos = pos; Page No -

```

```

while ((ch >= '0' && ch <= '9') || ch == '.') nextChar();
x = Double.parseDouble(exp.substring(startPos, pos));

```

```
return x;
} }.parse();
}}
```

**** Practical: 08 Title: Create an application for demonstration of Scroll view in android.**

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    >
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical">
        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Scroll....."/>
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="April Fool:)"
            android:layout_gravity="center_horizontal"
            android:layout_marginTop="500pt"
        />
    </LinearLayout>
</ScrollView>
```

****Practical: 09 Title: Demonstrate use of intent in android**

MainActivity.java

```

package com.example.intentdemo;

import android.content.Intent;

import android.net.Uri;

import android.os.Bundle;

import android.view.View;

import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnExplicit, btnImplicit;

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        btnExplicit = findViewById(R.id.btnExplicit);

        btnImplicit = findViewById(R.id.btnImplicit);

        // Explicit Intent (Open Second Activity)

        btnExplicit.setOnClickListener(new View.OnClickListener() {

            @Override

            public void onClick(View v) {

                Intent intent = new Intent(MainActivity.this, SecondActivity.class);

                intent.putExtra("message", "Welcome to Second Activity");

                startActivity(intent);

            }

        });

        // Implicit Intent (Open Web Browser)

        btnImplicit.setOnClickListener(new View.OnClickListener() {

            @Override

            public void onClick(View v) {

                Intent intent = new Intent(Intent.ACTION_VIEW);

                intent.setData(Uri.parse("https://www.google.com"));

                startActivity(intent);

```

```
}});}}
```

SecondActivity.java

```
package com.example.intentdemo;

import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {
    TextView textView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
        textView = findViewById(R.id.textView);
        String msg = getIntent().getStringExtra("message");
        textView.setText(msg);
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <Button
        android:id="@+id/btnExplicit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Open Second Activity"/>

    <Button
        android:id="@+id/btnImplicit"
```



```

    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Open Google"
    android:layout_marginTop="20dp"/>
</LinearLayout>

activity_second.xml

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">
    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp"/>
    </LinearLayout>

```

**** Practical: 10 Title: Create application to demonstrate menu option.**

MainActivity.java

```

package com.example.menudemo;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

```

```

setContentView(R.layout.activity_main);
}

// Create Menu

@Override

public boolean onCreateOptionsMenu(Menu menu) {
    getMenuInflater().inflate(R.menu.menu_main, menu);
    return true;
}

// Handle Menu Click

@Override

public boolean onOptionsItemSelected(MenuItem item) {
    int id = item.getItemId();
    if (id == R.id.menu_settings) {
        Toast.makeText(this, "Settings Clicked", Toast.LENGTH_SHORT).show();
        return true;
    }
    if (id == R.id.menu_about) {
        Toast.makeText(this, "About Clicked", Toast.LENGTH_SHORT).show();
        return true;
    }
    if (id == R.id.menu_exit) {
        finish();
        return true;
    }
    return super.onOptionsItemSelected(item);
}}

```

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

xmlns:android="http://schemas.android.com/apk/res/android"

android:layout_width="match_parent"

```

```
android:layout_height="match_parent"
android:gravity="center">
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Menu Demonstration App"
android:textSize="20sp"/>
</LinearLayout>
```

Create Menu Folder

Right click → res → New → Android Resource Directory →

Select Resource type = menu

menu_main.xml (res/menu/menu_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
<item
android:id="@+id/menu_settings"
android:title="Settings"/>
<item
android:id="@+id/menu_about"
android:title="About"/>
<item
android:id="@+id/menu_exit"
android:title="Exit"/>
</menu>
```

****Practical: 11 Title: .Create application to demonstrate progress bar.**

MainActivity.java

```
package com.example.progressdemo;

import android.os.Bundle;
import android.os.Handler;
import android.widget.Button;
```

```

import android.widget.ProgressBar;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ProgressBar progressBar;

    Button btnStart;

    int progressStatus = 0;

    Handler handler = new Handler();

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        progressBar = findViewById(R.id.progressBar);

        btnStart = findViewById(R.id.btnStart);

        btnStart.setOnClickListener(v -> {

            progressStatus = 0;

            progressBar.setProgress(progressStatus);

            new Thread(() -> {

                while (progressStatus < 100) {

                    progressStatus += 10;

                    handler.post(() -> progressBar.setProgress(progressStatus));

                    try {

                        Thread.sleep(500);

                    } catch (InterruptedException e) {

                        e.printStackTrace();

                    }

                }

            }).start();

        });

    }

```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:gravity="center"
android:padding="20dp">
<ProgressBar
android:id="@+id/progressBar"
style="?android:attr/progressBarStyleHorizontal"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:max="100"/>
<Button
android:id="@+id/btnStart"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Start Progress"
android:layout_marginTop="20dp"/>
</LinearLayout>
```